

OSAN AB, Korea, Republic of

WMO: 471220

Lat: 37.091N

Lon: 127.030E

Elev: 12

StdP: 101.19

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.1	-11.1	-18.8	0.7	-8.7	-16.8	0.9	-7.5	9.7	-1.9	8.7	-2.2	2.1	70	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	33.4	26.0	32.0	25.4	30.8	24.7	27.8	30.1	26.9	29.8	26.2	29.0	3.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	28.4	26.2	21.6	27.9	25.2	20.4	27.3	89.0	31.4	85.0	29.5	81.6	28.7	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.3	6.3	-16.2	34.9	2.5	1.5	-18.0	36.0	-19.4	36.9	-20.8	37.7	-22.7	38.8
			-16.6	28.9	2.3	1.3	-18.3	29.8	-19.6	30.6	-20.9	31.3	-22.6	32.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	-2.6	0.0	5.4	11.9	17.8	22.5	25.9	26.4	21.3	14.4	6.9	-0.3	(6)
(7)		DBStd	10.60	4.10	3.96	3.60	3.58	2.69	2.11	2.28	2.58	2.72	3.48	4.24	4.20	(7)
(8)		HDD10.0	1275	391	282	147	19	0	0	0	0	7	110	319		(8)
(9)		HDD18.3	2849	649	514	399	193	42	1	0	6	126	342	577		(9)
(10)		CDD10.0	2204	0	1	6	78	242	375	492	509	338	145	18	0	(10)
(11)		CDD18.3	738	0	0	0	2	26	126	234	251	94	5	0	0	(11)
(12)		CDH23.3	6157	0	0	0	33	251	856	2034	2358	586	38	1	0	(12)
(13)		CDH26.7	2014	0	0	0	3	44	226	703	911	126	1	0	0	(13)
(14)	Wind	WSAvg	2.5	2.5	2.7	3.1	3.0	2.8	2.4	2.5	2.4	2.1	2.2	2.4	2.4	(14)
(15)	Precipitation	PrecAvg	1372	19	29	44	82	94	144	387	289	168	48	44	25	(15)
(16)		PrecMax	2138	63	78	130	189	240	432	877	880	586	185	128	58	(16)
(17)		PrecMin	802	1	1	4	20	12	29	126	61	20	1	7	4	(17)
(18)		PrecStd	312	15	21	29	42	45	93	174	190	147	39	26	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	15.1	20.1	26.9	30.1	32.1	35.1	31.1	26.2	20.9	13.1		(19)
(20)		MCWB	6.7	10.6	12.0	17.5	17.3	21.5	25.7	26.5	23.2	18.4	15.0	9.2		(20)
(21)		2%	DB	7.8	11.2	17.1	23.9	27.9	30.2	33.1	34.1	29.5	24.2	18.2	10.2	(21)
(22)		MCWB	4.7	6.6	10.2	14.7	18.0	21.2	26.0	26.2	22.4	17.4	13.2	7.0		(22)
(23)		5%	DB	5.8	9.0	15.0	21.8	26.0	29.1	31.8	32.8	28.1	22.8	16.2	8.3	(23)
(24)		MCWB	3.1	4.9	8.9	13.5	17.2	20.9	25.6	26.1	21.7	16.7	12.1	5.8		(24)
(25)		10%	DB	4.0	6.9	12.8	19.8	24.2	27.9	30.2	31.1	26.8	21.0	14.2	6.8	(25)
(26)		MCWB	1.8	3.7	7.7	12.6	16.8	20.8	25.1	25.7	20.8	16.1	11.0	4.6		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	11.8	14.2	19.3	22.3	26.1	28.9	29.9	26.8	21.2	17.2	10.1	(27)
(28)		MDCB	9.3	13.7	17.0	23.3	24.7	27.3	30.7	31.0	28.4	23.5	18.8	11.8		(28)
(29)		2%	WB	5.1	7.4	11.9	16.7	20.6	24.3	27.5	28.1	24.9	19.6	14.7	7.9	(29)
(30)		MDCB	6.9	10.1	15.2	21.4	23.7	26.4	29.8	30.4	26.8	21.9	16.8	9.5		(30)
(31)		5%	WB	3.5	5.5	10.0	15.0	19.3	23.3	26.8	27.2	23.6	18.2	13.0	6.2	(31)
(32)		MDCB	5.0	8.1	13.2	19.3	23.0	25.9	29.4	30.3	25.9	20.7	15.2	7.9		(32)
(33)		10%	WB	2.0	4.0	8.3	13.5	18.2	22.4	26.1	26.5	22.5	17.0	11.4	4.7	(33)
(34)		MDCB	3.4	6.2	11.7	17.7	22.1	25.2	28.8	29.6	24.8	19.8	13.7	6.4		(34)
(35)	Mean Daily Temperature Range	MDBR	10.5	11.4	12.0	12.9	11.8	9.9	7.5	7.9	9.8	11.9	11.1	10.3		(35)
(36)		5% DB	MCDBR	11.5	14.1	15.5	16.1	14.7	12.1	9.5	9.8	11.0	13.1	13.2	12.1	(36)
(37)		MCWBR	8.5	9.8	9.2	8.2	6.4	5.2	4.0	4.0	5.2	7.0	8.8	9.1		(37)
(38)		5% WB	MCDBR	10.3	12.0	13.1	13.1	11.3	9.2	7.7	8.2	8.7	10.8	11.2	11.3	(38)
(39)		MCWBR	8.4	9.2	9.1	7.8	6.5	5.3	4.3	4.3	5.1	7.2	8.5	9.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.443	0.491	0.574	0.611	0.625	0.697	0.658	0.582	0.509	0.497	0.480	0.439		(40)
(41)		taud	1.942	1.856	1.710	1.677	1.697	1.618	1.747	1.925	2.051	1.958	1.943	1.967		(41)
(42)		Ebn at Noon	694	712	693	699	699	649	672	715	745	700	649	663		(42)
(43)		Edh at Noon	141	174	220	240	239	259	226	185	155	155	140	129		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.48	3.25	4.29	4.99	5.59	5.18	4.14	4.37	4.11	3.59	2.46	2.18		(44)
(45)		RadStd	0.15	0.27	0.43	0.37	0.52	0.51	0.61	0.63	0.45	0.27	0.26	0.15		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page